ASEAN GEOAI FUSION 2026

Universiti Kebangsaan Malaysia
The National University of Malaysia

AGAIF 2026 BOOTCAMP 1: FOUNDATIONAL GEOAI

Geospatial Data Sources
Session 4

Online
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GEOSPATIAL DATA SOURCES

OpenStreetMap, Government Open Data Portals,
Telecommunications Infrastructure Datasets & Socio-economic Data Integration



01 OPENSTREETMAP & VOLUNTEER GEOGRAPHIC INFORMATION (VGI)

OpenStreetMap
The free and open map of the world, created by volunteers.

DATA TYPES
- Roads & Railways
- Buildings
- Points of Interest (POI)
- Land Use & Natural Features
- Water Bodies

ADVANTAGES
- ✓ Free & open
- ✓ Global coverage
- ✓ Frequently updated
- ✓ Local knowledge

LIMITATIONS
- ⚠ Variable data quality
- ⚠ Incomplete in some areas
- ⚠ Not official / not authoritative

02 GOVERNMENT OPEN DATA PORTALS

Provide authoritative and reliable geospatial datasets for public use.

EXAMPLES IN ASIA

COUNTRY	PORTAL	KEY GEOSPATIAL DATA
Malaysia	data.gov.my	Census, Roads, Health, Education, Weather
Singapore	data.gov.sg	Buildings, Transport, Environment, Population
Indonesia	data.go.id	Population, Disaster, Agriculture, Transport
Thailand	data.go.th	Population, Tourism, Health, Roads
Philippines	data.gov.ph	Disaster, Census, Health, Education
Vietnam	data.gov.vn	Boundaries, Transport, Environment, Climate
India	data.gov.in	Census, Railways, Weather, Agriculture
Japan	data.go.jp	Disaster Risk, Transport, Population, Weather
South Korea	data.go.kr	Buildings, Transport, Environment, Health
Taiwan	data.gov.tw	Land Use, Transport, Health, Tourism
China	data.gov.cn	Transport, Economy, Environment, Weather*

*Some datasets may require registration or have usage restrictions.

03 TELECOMMUNICATIONS INFRASTRUCTURE DATASETS

Spatial data about communication networks and digital infrastructure.

DATA EXAMPLES
- Mobile towers (2G/3G/4G/5G)
- Fibre optic networks
- Microwave links
- Internet exchange points
- Wi-Fi hotspots & small cells
- Satellite ground stations

SOURCES
- National telecom regulators
- Telecom operators
- OpenCellID, GSMA Open Data
- ITU, Ookla Open Data, Mozilla Location Service

APPLICATIONS
Network Planning · Smart Cities · Disaster Response · Mobility Analysis

04 SOCIO-ECONOMIC DATA INTEGRATION

Integrating socio-economic data with spatial data provides deeper insights for decision-making.

DATA EXAMPLES
- Population & Demographics
- Income & Poverty
- Education
- Health
- Employment
- Housing
- Economic Activity

SOURCES
- National Census & Surveys
- World Bank Open Data
- UNData, ADB, Asian Development Bank
- National Statistical Offices

INTEGRATION EXAMPLE
Spatial Data (Flood Hazard, Roads, Buildings) + Socio-economic Data (Population, Income, Vulnerability) = Flood Vulnerability Map

LEARNING OUTCOMES

At the end of this lecture, students will be able to:

- Explain the major sources of geospatial data.
- Identify reliable open geospatial datasets available in Asia.
- Compare volunteered and authoritative datasets.
- Understand socio-economic data integration in GIS.
- Select appropriate datasets for GIS applications.

GIS APPLICATIONS SUPPORTED

Urban Planning · Disaster Management · Transportation · Public Health · Environmental Monitoring · Agriculture · Climate Change · Sustainable Development (SDGs)

BEST PRACTICES

- ✓ Check accuracy, completeness & update frequency
- ✓ Read metadata and understand data limitations
- ✓ Ensure proper coordinate system & scale
- ✓ Respect licensing and data usage policies
- ✓ Integrate multiple datasets for better insights

KEY MESSAGE

Multiple geospatial data sources complement each other. Integrating open, authoritative, infrastructural and socio-economic datasets leads to better insights, smarter decisions and a more sustainable future.

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WHY GEOSPATIAL DATA SOURCES MATTER

Geospatial data supports a wide range of applications that improve decision-making, enhance efficiency and build a better, more sustainable future.

URBAN PLANNING



Supports land use planning, infrastructure development, zoning and sustainable urban growth.

SMART CITIES



Enables data-driven urban management, real-time monitoring and improved citizen services.

DISASTER RISK REDUCTION



Helps in hazard mapping, early warning, risk assessment and emergency response.

AGRICULTURE



Supports precision farming, crop monitoring, soil analysis and resource optimization.



GEOSPATIAL DATA

The foundation for smarter decisions and a better world.

ENVIRONMENTAL MONITORING



Tracks changes in land cover, water resources, forests and biodiversity for better environmental management.

PUBLIC HEALTH



Helps map disease outbreaks, health facilities, population distribution and access to healthcare.

TRANSPORTATION



Improves route planning, traffic management, infrastructure design and public transport efficiency.

CLIMATE CHANGE



Monitors climate patterns, sea level rise, extreme weather and supports adaptation and mitigation strategies.

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

Provides essential spatial data and indicators to monitor progress, ensure inclusivity and achieve a sustainable future for all.

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TYPES OF GEOSPATIAL DATA SOURCES

Geospatial data comes from diverse sources. Each type plays a unique role in building accurate maps, enabling analysis and supporting better decision-making.



01 COMMUNITY-BASED

Example:
OpenStreetMap



02 GOVERNMENT

Example:
National Geoportals



03 COMMERCIAL

Example:
Google Maps API



04 REMOTE SENSING

Example:
Landsat, Sentinel



05 TELECOMMUNICATION

Example:
Cell towers,
Fibre network



06 SOCIO-ECONOMIC

Example:
Census,
Household surveys





Why it matters?

Integrating multiple geospatial data sources provides a more comprehensive view of our world and supports smarter, data-driven solutions.



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OPENSTREETMAP (OSM)

The Free Map of the World, Built by People Like You.

OpenStreetMap (OSM) is a global, collaborative mapping project that creates and provides free geographic data for anyone to use.

OpenStreetMap
The World is Mapped by You.

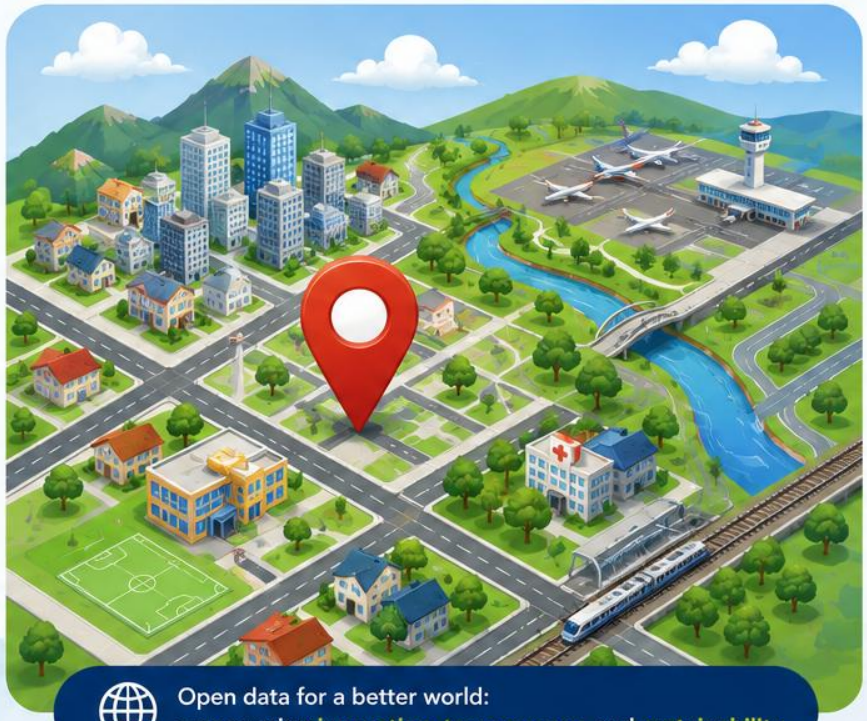
WEBSITE
https://www.openstreetmap.org

Join the community.
Map the world.
Make a difference!

OPENSTREETMAP IS

- **Free**
  No cost to use for any purpose.
- **Open-source**
  Free license, open data, open possibilities.
- **Community maintained**
  Thousands of volunteers contribute and improve the map every day.
- **Global**
  Mapping the whole world, for everyone.
- **Continuously updated**
  Changes on the ground are reflected quickly.





Open data for a better world:
empowering innovation, transparency, and sustainability.

CONTAINS

- **Roads**
  Streets, highways, paths and more.
- **Buildings**
  Residential, commercial, industrial and more.
- **Rivers**
  Rivers, streams, canals and lakes.
- **Land use**
  Parks, forests, farmland, residential areas and more.
- **Schools**
  Educational institutions of all levels.
- **Hospitals**
  Hospitals, clinics and healthcare facilities.
- **Parks**
  Parks, gardens and recreation areas.
- **Railways**
  Rail lines, stations and rail infrastructure.
- **Airports**
  Airports, runways and air navigation facilities.

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VOLUNTEER GEOGRAPHIC INFORMATION (VGI)

People. Places. Purpose.
Crowdsourcing a Better World.

DEFINITION

Geographic information **voluntarily** collected by individuals.

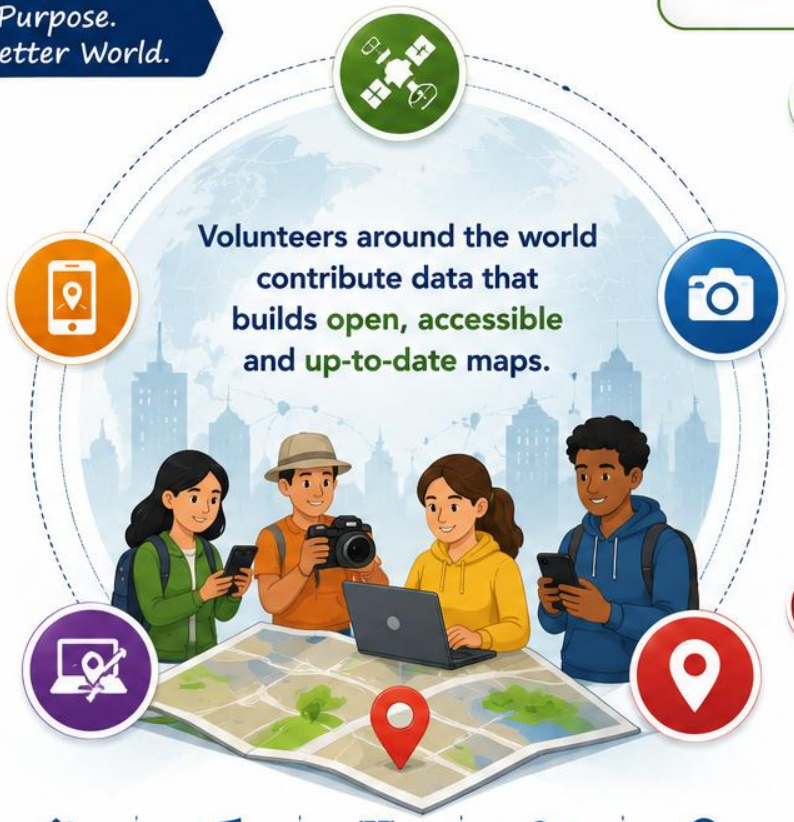
EXAMPLES

OpenStreetMap
A free, editable map of the world created by volunteers.

Mapillary
Crowdsourced street-level photos to map the world.

Humanitarian OpenStreetMap Team (HOT)
Mapping for humanitarian response and disaster relief.

Wikimapia
A collaborative map with place information from people around the world.



Disaster Response | Environmental Monitoring | Infrastructure Development | Community Engagement | Education & Research

ADVANTAGES

Rapid updates
Information is updated quickly by a large number of contributors.

Local knowledge
Volunteers provide on-the-ground insights and local context.

Low cost
Free data reduces mapping and surveying expenses.

Excellent disaster support
Fast mapping support during emergencies and disasters.

LIMITATIONS

Variable quality
Data accuracy and completeness may vary.

Incomplete rural coverage
Less contributor activity in remote or rural areas.

Volunteer bias
Data may reflect the interests or perspectives of contributors.

Together, volunteers create a **stronger, smarter** and **more connected** world through geographic information.

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GOVERNMENT OPEN DATA PORTALS

Open Data. Transparent Government. Smarter Decisions.

Governments provide **authoritative** and **high-quality** geospatial datasets to support transparency, innovation and public value.

AUTHORITATIVE DATASETS INCLUDE

- **Administrative Boundaries**
  Country, state, district and local government boundaries.
- **Transportation**
  Roads, railways, airports, ports and public transport.
- **Population**
  Census data, demographics and population statistics.
- **Buildings**
  Building footprints, public facilities and infrastructure.
- **Environment**
  Rivers, lakes, forests, protected areas, climate and air quality.
- **Land Use**
  Land cover, land use, zoning and spatial planning.
- **Agriculture**
  Crop data, farms, soil, fisheries and livestock.
- **Health**
  Hospitals, clinics, health facilities and health indicators.
- **Education**
  Schools, universities, enrolment and education statistics.
- **Disaster Management**
  Hazard maps, risk zones, early warning data and emergency facilities.



OPEN ACCESS
Free and easy access for everyone.

INTEROPERABLE
Data in standard formats for easy use.

TRANSPARENT
Strengthening trust and accountability.

BENEFITS

RELIABLE
Authoritative data from official government sources.

STANDARDIZED
Consistent formats, definitions and classification systems.

FREQUENTLY MAINTAINED
Regular updates ensure timely and current information.

OFFICIAL STATISTICS
Data based on official surveys, censuses and administrative records.

Open government data **empowers** research, innovation, business and **better public services**.

Open data is the foundation for **smarter cities**, **stronger communities** and a **sustainable future**.

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GOVERNMENT OPEN DATA PORTALS IN ASIA

Open Data. Open Opportunities.
Building a **Transparent** and **Data-Driven** Asia.



COUNTRY	PORTAL	MAIN AGENCY
Malaysia	data.gov.my	Ministry of Digital
Singapore	data.gov.sg	Government Technology Agency (GovTech)
Indonesia	data.go.id	Bappenas
Thailand	data.go.th	Digital Government Development Agency
Philippines	data.gov.ph	Department of Information and Communications Technology
Vietnam	data.gov.vn	Ministry of Information & Communications
India	data.gov.in	National Informatics Centre
Japan	data.go.jp	Digital Agency
South Korea	data.go.kr	Ministry of the Interior and Safety
Taiwan	data.gov.tw	National Development Council

WHY IT MATTERS

These portals provide free and authoritative data that empowers citizens, researchers, businesses and governments to **innovate**, **analyze** and create **impactful solutions**.

- Research & Innovation
- Policy Making
- Smart Cities
- Sustainable Development
- Disaster Management

One Region. Many Portals. Shared Vision.
OPEN DATA FOR A BETTER ASIA.

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MALAYSIA

OPEN DATA FOR A SMARTER NATION

Reliable. Accessible. Up-to-date.
Explore trusted geospatial data from Malaysia's official portals.



data.gov.my
THE NATIONAL OPEN DATA PORTAL

One-stop platform for accessing a wide range of open datasets from Malaysian government agencies.

AVAILABLE DATASETS

Population
Population estimates, demographics and statistics.

Census
Census data and socio-economic information.

Roads
Road network, highways and infrastructure.

Schools
Location of schools and educational facilities.

Healthcare
Hospitals, clinics and health facilities.

Economy
Economic indicators, business and trade data.

Weather
Weather observations, forecasts and climate data.

Crime
Crime statistics and public safety information.

Tourism
Tourist attractions, accommodations and statistics.

Website: https://data.gov.my
Empowering decisions through open and transparent data.

MyGeoPortal (JUPEM)
OFFICIAL GEOSPATIAL DATA PORTAL

Authoritative geospatial data provided by JUPEM for mapping, planning and development.

AVAILABLE DATASETS

Administrative Boundaries
National, state, district and local government boundaries.



Topographic Maps
Topographic maps at various scales for detailed planning.



Hydrography
Rivers, lakes, coastlines and water features.



Geodetic Control
Geodetic reference stations and control points.



Land Use
Land use and land cover information.



Transportation
Roads, railways, airports and other transport networks.



Website: https://www.mygeoportal.gov.my
Building Malaysia with accurate and reliable geospatial information.

Two trusted portals. Countless opportunities.
DATA THAT DRIVES DEVELOPMENT.

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SINGAPORE

OPEN DATA. SMART NATION.

Access authoritative geospatial data to support innovation, planning and better decisions.



Data.gov.sg

THE NATIONAL OPEN DATA PORTAL

Singapore's official portal for open government data from public agencies.

PROVIDES

- **Building Footprints** — Geospatial outlines of buildings across Singapore.
- **MRT Stations** — Locations and details of all MRT stations.
- **Bus Routes** — Bus route networks, stops and schedules.
- **Population** — Population estimates and demographic information.
- **Housing** — Public and private housing data and statistics.
- **Environment** — Environmental quality, greenery, and conservation data.
- **Weather** — Weather observations, forecasts and climate data.



BENEFITS

- **Reliable** — Authoritative and high-quality data
- **Accessible** — Free and easy to access
- **Up-to-date** — Regularly updated datasets
- **Innovative** — Powering research, apps and smart solutions



OneMap

The official map of Singapore by the Singapore Land Authority.

PROVIDES

- **Parcel Boundaries** — Land parcel outlines and property boundaries.
- **Buildings** — Building footprints and building information.
- **Roads** — Road networks and road information.
- **Schools** — Location of schools and education facilities.
- **Healthcare** — Location of hospitals and healthcare facilities.
- **Elevation** — Terrain elevation and contour information.
- **Planning Boundaries** — Development and planning boundaries.



OPEN DATA. SMARTER PLANS. BETTER LIVES.

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OPEN DATA PORTALS

INDONESIA • THAILAND • PHILIPPINES

Open Data. Stronger Communities. Smarter Future.

INDONESIA

OPEN DATA PORTALS

SATU DATA INDONESIA
data.go.id

INA-GEOPORTAL
tanahair.indonesia.go.id

DATASETS

- **DEM** — Digital Elevation Model data
- **Forest** — Forest cover and resources data
- **Disaster** — Disaster risk and events data
- **Agriculture** — Agricultural land and production data
- **Transportation** — Transportation networks and infrastructure

THAILAND

data.go.th
Thailand Open Data Portal

DATASETS

- **Population** — Population and demographic data
- **Roads** — Road networks and spatial data
- **Tourism** — Tourism attractions and statistics
- **Healthcare** — Health facilities and services data
- **Agriculture** — Agricultural statistics and land use

PHILIPPINES

data.gov.ph
Philippines Open Data Portal

DATASETS

- **Disaster** — Disaster risk and alerts data
- **Census** — Census and population data
- **Roads** — Road networks and infrastructure
- **Health** — Health statistics and facilities
- **Education** — Education and schools data

OPEN DATA BENEFITS EVERYONE

- Transparency and Accountability
- Better Decisions and Services
- Innovation and Economic Growth
- Stronger Communities and Collaboration

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RECOMMENDED ASIAN GOVERNMENT OPEN DATA PORTALS

Trusted • Open • Geospatial • Actionable

Access reliable geospatial and socio-economic data to support evidence-based decisions, planning and innovation.



COUNTRY	PORTAL	KEY GEOSPATIAL DATA AVAILABLE
Malaysia	https://data.gov.my	Census, roads, schools, health, weather, economy
Malaysia (JUPEM)	https://www.mygeoportal.gov.my	Administrative boundaries, topographic maps, hydrography, land use
Singapore	https://data.gov.sg	Buildings, transport, housing, environment, weather
Singapore (OneMap)	https://www.onemap.gov.sg	Parcels, roads, buildings, planning, healthcare
Indonesia	https://data.go.id	Population, agriculture, disaster, transportation
Indonesia (Ina-Geoportal)	https://tanahair.indonesia.go.id	Base maps, DEM, land cover, rivers, roads
Thailand	https://data.go.th	Population, tourism, healthcare, roads
Philippines	https://data.gov.ph	Disaster, census, education, health
Vietnam	https://data.gov.vn	Administrative boundaries, transport, environment

WHY IT MATTERS

- Better Spatial Insights
- Evidence-Based Decisions
- Inclusive Development
- Stronger Resilience

Open data. Smarter decisions. Better future for Asia.

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TELECOMMUNICATIONS INFRASTRUCTURE DATASETS

Connecting Data. Connecting People. Connecting the Future.

Telecommunications infrastructure datasets provide critical information for **connectivity** planning, digital inclusion, **smart cities** and **innovation**.

EXAMPLES

Mobile Towers
Location and attributes of mobile base stations.

4G / 5G Coverage
Network coverage areas and signal strength.

Fibre Optic Network
Routes and availability of fibre infrastructure.

Microwave Links
Point-to-point links for backhaul connectivity.

Internet Exchange
Locations of Internet Exchange Points (IXP).

Wi-Fi Hotspots
Public Wi-Fi locations and access points.



- Better Planning
- Digital Inclusion
- Smart Cities
- Resilience & Disaster Response
- Data-Driven Decisions

SOURCES

National Telecommunications Regulators
Official data from national regulatory bodies and authorities.

Mobile Operators
Network infrastructure data from telecom service providers.

OpenCellID
Crowdsourced cell tower and network information.

GSMA
Global mobile industry data and market intelligence.

ITU
International Telecommunication Union datasets and statistics.

Ookla Open Data
Internet performance and speed test data.

High-quality telecommunications data powers a connected and inclusive digital future.

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GIS APPLICATIONS OF TELECOMMUNICATION DATA

Turning Connectivity Data into Smarter Decisions



APPLICATIONS

NETWORK PLANNING
Optimize cell tower placement, coverage and capacity.

SMART CITIES
Support urban development, digital services and resource management.

TRAFFIC MONITORING
Analyze congestion patterns and improve traffic flow.

HUMAN MOBILITY
Understand movement patterns and population dynamics.

EMERGENCY RESPONSE
Locate affected areas and coordinate relief operations.

DISEASE SPREAD MODELLING
Model and predict the spread of infectious diseases.

INFRASTRUCTURE RESILIENCE
Assess risks and strengthen the resilience of critical infrastructure.

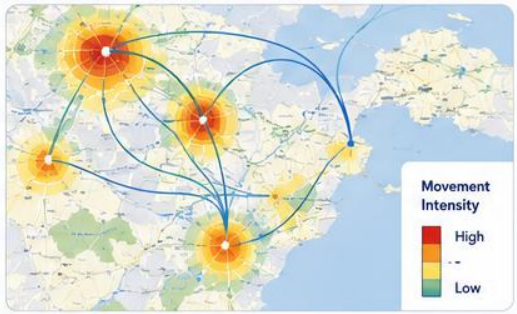


Spatial Analysis → Pattern Detection → Behaviour Insights → Predictive Modelling → Better Decisions

EXAMPLE

COVID-19 MOBILITY ANALYSIS

USING ANONYMIZED MOBILE PHONE DATA



- Track population movement between regions.
- Identify hotspots and high-risk areas.
- Support public health decisions and policy making.
- Improve preparedness and response to future outbreaks.

Telecommunication data + GIS technology = Smarter insights for a connected world.

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SOCIO-ECONOMIC DATA INTEGRATION

Integrating People, Economy and Places for Better Decisions

Combining socio-economic data with geospatial insights to understand communities, reduce inequality and support sustainable development.



SOCIO-ECONOMIC DATASETS INCLUDE

Population
Number and distribution of people in an area.

Household Income
Income levels and economic well-being of households.

Poverty
Poverty levels and vulnerability indicators.

Employment
Employment status, occupation and labour force data.

Education
School enrollment, literacy rates and education attainment.

Healthcare
Access to healthcare facilities and health indicators.

Housing
Housing conditions, ownership and infrastructure.

Economic Activity
Business activity, industry output and trade information.

Demographics
Age, gender, ethnicity and other demographic characteristics.

INTEGRATION CREATES POWERFUL INSIGHTS



SOURCES

National Census
Official population and housing census data.

Household Survey
Detailed data on household characteristics and welfare.

Labour Survey
Information on employment, wages and working conditions.

World Bank
Global development indicators and socio-economic data.

UNData
Data from United Nations agencies and programs.

Asian Development Bank
Economic and social data for Asia and the Pacific.

Integrated Data
Combines multiple socio-economic datasets.

Smarter Analysis
Reveals patterns and hidden relationships.

Better Decisions
Supports policies that improve lives.

Stronger Communities
Building an inclusive and resilient society.

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WHY INTEGRATE SOCIO-ECONOMIC DATA?

Combining spatial and socio-economic data allows us to answer real-world questions and make better decisions.

SPATIAL DATA
Maps, boundaries, infrastructure, environment

SOCIO-ECONOMIC DATA
Population, income, education, health, employment, etc.

INTEGRATION
Linking people, places and services for deeper insights

BETTER DECISIONS
Evidence-based planning for a better future

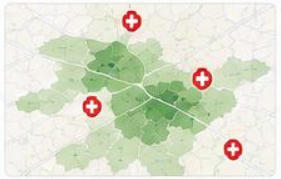


INTEGRATED INSIGHTS FOR REAL IMPACT

Where are vulnerable communities?



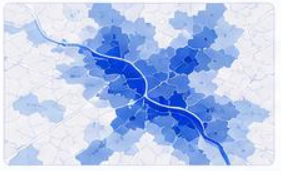
Which areas lack healthcare access?



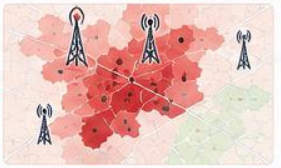
Where should new schools be built?



Which regions require flood mitigation?



Which communities have poor internet access?



Integrated socio-economic data turns information into action — creating inclusive, resilient and sustainable communities.

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EXAMPLE GIS INTEGRATION

CASE STUDY FLOOD VULNERABILITY ASSESSMENT

Integrating spatial and socio-economic data to identify areas and communities most at risk from flooding and support better decision-making.



1 SPATIAL DATASETS

Flood Hazard
Flood depth, extent, and frequency

Roads
Road network and access routes

Buildings
Building footprints and infrastructure

Rivers
Rivers, streams and waterways



2 SOCIO-ECONOMIC DATASETS

Population Density
Number of people per area

Income
Household income levels

Elderly Population
Proportion of elderly residents

Disability Statistics
Proportion of people with disabilities



3 OUTPUT

FLOOD VULNERABILITY MAP



Vulnerability Level
- Very High
- High
- Moderate
- Low
- Very Low

- Identify high-risk areas and communities
- Prioritize resources and interventions
- Support urban planning and zoning
- Improve disaster preparedness and resilience

THE POWER OF INTEGRATION

- Better Risk Identification
- Evidence-Based Decisions
- Protect Vulnerable Communities
- Smarter, Resilient Communities

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BEST PRACTICES FOR SELECTING DATA

The right data leads to reliable insights and better decisions.



01 ACCURACY

Choose data with high positional and attribute accuracy to ensure reliable analysis.

02 COMPLETENESS

Ensure the dataset is complete enough for your intended analysis and area of interest.

03 UPDATE FREQUENCY

Use data that is updated regularly to reflect the current real-world conditions.

04 METADATA

Review metadata to understand the data's origin, collection methods, and limitations.

05 COORDINATE REFERENCE SYSTEM

Ensure the data uses an appropriate and consistent CRS for accurate overlay and spatial analysis.

LICENSING

Check the license terms to ensure the data can be legally used and shared.

RELIABILITY

Choose data from trustworthy sources with a proven track record and credibility.

08 SCALE

Select data at the right scale and resolution for your analysis needs.

09 INTEROPERABILITY

Use data in standard formats that can easily integrate with other datasets and systems.

Careful data selection saves time, reduces risk, and delivers stronger, data-driven outcomes.

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SUMMARY

Building a Stronger Future with Integrated Geospatial Data

Geospatial information comes from multiple **complementary sources.**

1 OpenStreetMap

provides rapidly updated community mapping.



2 Government Open Data

offers authoritative datasets.





3 Telecommunications Data

supports smart cities and mobility analysis.



4 Socio-economic Information

explains human dimensions.



5 Integrating Multiple Datasets

produces better spatial decision-making.



INTEGRATE. ANALYZE. ACT.

Bringing data together to create **smarter**, more **inclusive**, and **resilient** communities.

Better Understanding → Smarter Decisions → Stronger Communities → Resilient Future

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